When Does Adaptive Alpha Control Help? A Controlled Comparison of Adversarial Debiasing and Post-Processing Calibration

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Abstract

In adversarial debiasing, a single hand-picked weight (α) sets the accuracy-fairness trade-off for the whole training run. We propose Adaptive Alpha Control, a dual-ascent controller that adjusts two separate weights every epoch, one for an equal-opportunity constraint and one for an average-odds constraint. Each target is set as a fraction of the task's own measured baseline bias. On COMPAS (30 seeds, paired significance testing), the controller reaches a different point on the accuracy-fairness frontier from the best fixed alpha found by sweeping. Its accuracy is statistically indistinguishable, its equal-opportunity gap is significantly smaller ($p < 0.0001$) and its average-odds gap is significantly larger ($p = 0.027$), so neither method dominates the other. We then compare both against post-processing equalized-odds calibration (Hardt et al., 2016), which needs no adversary or retraining. On COMPAS it has the lowest equal-opportunity and average-odds gaps of any method we tested and beats our adaptive method on all three metrics, although fixed $\alpha = 0.5$ and $\alpha = 1.0$ keep higher accuracy. The result does not carry over to a smaller second dataset. On German Credit (200 test rows against COMPAS's 1,056), post-processing becomes the worst method, significantly so on the average-odds gap, while every adversarial method, including ours, stays stable across both datasets. The paper's contribution is a controlled comparison that identifies the condition (available calibration-set size) under which each approach should be preferred.

1 Introduction

Adversarial debiasing [Zhang et al., 2018] trains a predictor against an adversary that tries to recover a sensitive attribute from the predictor's output. A single weight, α, sets how much the predictor listens to the adversary, and it is chosen by hand before training begins. The right value probably changes during a run. Early on, an undertrained predictor's output may carry little signal for the adversary to exploit whatever α is; later, as the predictor converges, bias can re-emerge in ways a value chosen at the start cannot anticipate. A static α is a compromise over the whole run, and finding even a reasonable value means sweeping several candidates and comparing their outcomes afterwards.

This paper replaces the fixed weight with a controller that adjusts two separate weights every epoch, one for each of two fairness criteria (equal opportunity and average odds). Each update depends on how far the current model is from a fairness target set relative to the task's own measured baseline bias. We frame the controller as dual ascent on a constrained optimization problem, which gives each design choice a justification: what the update does, why there are two weights instead of one, and why there is a safety floor on accuracy.

Once the controller was built, the next question was whether a fairness mechanism that intervenes during training is needed at all. We compare against post-processing equalized-odds calibration [Hardt et al., 2016], which needs no adversary and no retraining, only a calibration pass on an already-trained, unconstrained predictor. On our primary dataset (COMPAS), post-processing has the lowest fairness gaps of every method tested and beats our adaptive mechanism on all three metrics. We then test whether this holds on a second, smaller dataset (German Credit) and find that it does not. Post-processing depends on having enough held-out data to calibrate reliable per-group thresholds, a condition COMPAS meets and German Credit does not. The paper therefore does not rank the methods. It identifies the condition under which each approach is preferable.

Contributions:

1. Adaptive Alpha Control: a dual-ascent controller for the predictor/adversary weight in adversarial debiasing, with separate multipliers for the equal opportunity (DEO) and average-odds (DAO) gaps, each targeting a fraction of the task's own measured baseline bias.

2. A comparison against a fixed-alpha sweep (3 points: $\alpha \in \{0.5, 1.0, 2.0\}$) across 30 seeds, showing the adaptive method lands on or near the best fixed-alpha operating point without requiring the sweep.

3. A controlled comparison against post-processing equalized-odds calibration [Hardt et al., 2016]. On COMPAS it gives the lowest DEO and DAO of any method, but on German Credit, where the calibration set is small, it becomes unreliable on the compound metric (DAO).

4. Results on two datasets (COMPAS, German Credit) with paired significance testing (Wilcoxon signed-rank, Holm-corrected).

2 Related Work

Adversarial debiasing. Zhang et al. [2018] introduce a predictor/adversary minimax game where a predictor is trained to accomplish a task while an adversary tries to recover a sensitive attribute from only the predictor's output. The predictor's loss subtracts the adversary's (scaled by a weight α), so the predictor is pushed to make its output uninformative about the sensitive attribute to the extent α demands. In the original formulation α is fixed and chosen by hand; we replace it with a per-epoch controller.

Constrained optimization for fairness. Agarwal et al. [2018] and Cotter et al. [2019] treat a fairness gap as a constraint instead of a soft penalty term, using Lagrangian dual methods that target the constraint through alternating gradient ascent on a multiplier. Our controller belongs to the same family, applied inside a per-epoch training loop instead of as an outer reduction over a black-box classifier.

Post-processing. Hardt et al. [2016] show that, given an already-trained (unconstrained) classifier's scores, per-group decision thresholds can be chosen to satisfy equalized odds exactly while maximizing accuracy. Each group randomly mixes two thresholds, and the operating point is found by searching the intersection of the groups' ROC convex hulls. This needs no retraining and no adversary, but it does need enough held-out data per group to estimate a reliable ROC curve.

Domain-adversarial training. Ganin & Lempitsky [2015] introduce the gradient-reversal layer, which lets a minimax game of this kind be trained with a single joint backward pass instead of alternating updates. We use it as an implementation-level ablation (Section 6) to check whether our alternating-update implementation was costing anything relative to the more common DANN-style setup. It was not: the two gave almost the same results.

COMPAS and algorithmic recidivism prediction. Angwin et al. [2016] is the ProPublica investigation that reported COMPAS's racial disparity in false positive rates, a widely used real-world case for equalized-odds fairness criteria. Its data is the primary dataset in this paper.

3 Method

3.1 Predictor/adversary architecture

Predictor: Dense(32, relu) $\to$ Dense(16, relu) $\to$ Dense(1, sigmoid). Input is the full feature vector; output is a single predicted probability for the task label y.

Adversary: Dense(16, relu) $\to$ Dense(8, relu) $\to$ Dense(1, sigmoid). Its only input is the predictor's scalar output $\hat{y}$. It never sees the raw features or the true label, so it can only succeed if $\hat{y}$ itself leaks the sensitive attribute z.

3.2 Fairness metrics

Let TPR_g and FPR_g denote the true/false positive rate within sensitive-attribute group $g \in \{0, 1\}$.

$$\mathrm{DEO} = |\mathrm{TPR}_0 - \mathrm{TPR}_1| \tag{1}$$

$$\mathrm{DAO} = \tfrac{1}{2}\left(|\mathrm{TPR}_0 - \mathrm{TPR}_1| + |\mathrm{FPR}_0 - \mathrm{FPR}_1|\right) \tag{2}$$

Both are computed on held-out data. A value of 0 is perfectly fair by this definition, and higher values mean more bias.

3.3 Baseline adversarial debiasing (fixed alpha)

Following Zhang et al. [2018], for a fixed scalar α:

$$L_P = \mathrm{BCE}(y, \hat{y}) - \alpha \cdot \mathrm{BCE}(z, \hat{z}) \tag{3}$$

$$L_A = \mathrm{BCE}(z, \hat{z}) \tag{4}$$

For each minibatch, the adversary is updated first with the predictor's weights held fixed, then the predictor is updated with the adversary's weights held fixed. The alternation matters because the second term of L_P depends on the adversary's current best response. With a stale or simultaneously updating adversary, the signal to maximize the adversary's loss would be wrong.

3.4 Adaptive Alpha Control (dual ascent)

Reframe the problem as constrained optimization:

$$\min \quad \mathrm{BCE}(y, \hat{y}) \tag{5}$$

$$\text{s.t.} \quad \mathrm{DEO} \le \epsilon_{\mathrm{DEO}}, \quad \mathrm{DAO} \le \epsilon_{\mathrm{DAO}} \tag{6}$$

with two separate dual variables $\alpha_{\text{DEO}}, \alpha_{\text{DAO}} \geq 0$ updated by gradient ascent on the Lagrangian after every epoch:

$$\alpha_{\text{DEO}} \leftarrow \max(0,\ \alpha_{\text{DEO}} + \eta \cdot (\text{DEO} - \epsilon_{\text{DEO}})) \tag{7}$$

$$\alpha_{\text{DAO}} \leftarrow \max(0,\ \alpha_{\text{DAO}} + \eta \cdot (\text{DAO} - \epsilon_{\text{DAO}})) \tag{8}$$

With a single shared adversary, the two multipliers would just add together and be redundant. To make them independent, the adversary is conditioned on the true label: two separate adversary heads are trained, one on the part of each batch where $y = 1$ (a direct proxy for the TPR gap, and so for DEO) and one where $y = 0$ (which adds the FPR-gap component that DAO needs):

$$L_P = \text{BCE}(y, \hat{y}) - \alpha_{\text{DEO}} \cdot L_{A,y=1} - \alpha_{\text{DAO}} \cdot \tfrac{1}{2}(L_{A,y=1} + L_{A,y=0}) \tag{9}$$

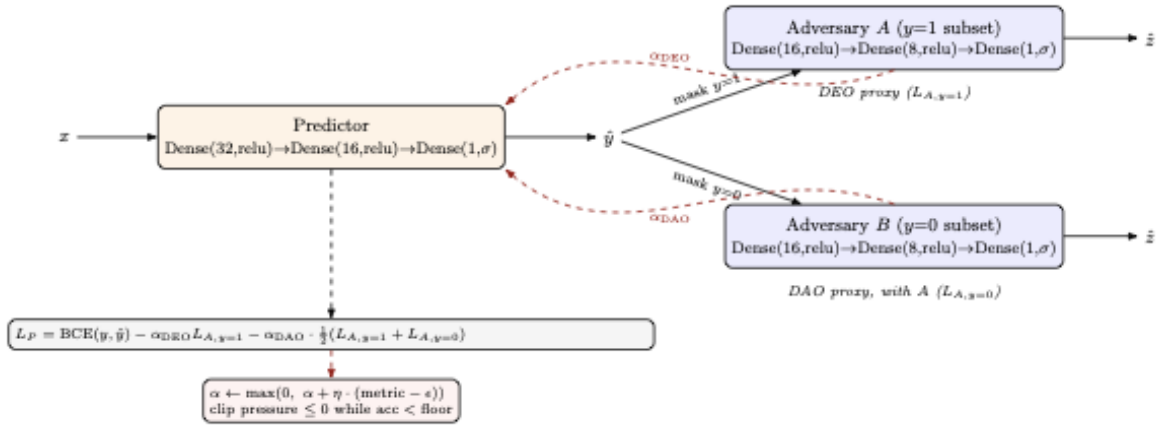


Figure 1: Adaptive Alpha Control: the predictor's output is split by true label into two subsets, each feeding a separate adversary head. Each head's loss proxies one fairness metric's leakage (DEO for the $y = 1$ subset, DAO for both); dual ascent on $\alpha_{\text{DEO}}, \alpha_{\text{DAO}}$ feeds back into the predictor's loss every epoch.

ϵ_{DEO} and ϵ_{DAO} are set to 30% of the dataset's own measured no-mitigation baseline DEO and DAO, so the target scales with how biased the dataset is to begin with. A hard safety clip stops the ascent step from increasing alpha while accuracy is below a floor (0.60 for COMPAS and 0.66 for German Credit, chosen after looking at the test accuracy that fixed $\alpha = 2.0$ reached; see Section 7). When accuracy is below the floor, the pressure term is clamped to be non-positive for that epoch, so alpha can still fall but cannot rise.

3.5 Post-processing equalized-odds calibration

Following Hardt et al. [2016]: train a plain unconstrained predictor (no adversary, full task accuracy). On held-out calibration data, compute each group's ROC curve and its upper concave hull (the set of (FPR, TPR) points achievable by randomly mixing two thresholds). Search for the (FPR*, TPR*) point that both groups' hulls can reach and that maximizes overall accuracy, then realize it per group by randomly mixing the two empirical thresholds on either side of that point on the group's hull. Evaluation uses a held-out split disjoint from the calibration split, to avoid optimistic bias.

4 Experimental Setup

4.1 Datasets

COMPAS [Angwin et al., 2016]: 7,214 rows, filtered following the original ProPublica analysis (screening within 30 days of arrest, valid recidivism data, non-traffic charges, a produced COMPAS score). We kept the two groups ProPublica's own comparison focuses on (African-American, Caucasian) to get a binary sensitive attribute. Target: two-year recidivism. After filtering: 5,278 rows (4,222 train / 1,056 test), 12 features.

German Credit: 1,000 rows, no filtering needed (800 train / 200 test, 56 features after one-hot encoding). Target: good vs. bad credit risk. Sensitive attribute: gender (binary in this dataset). We included it because it is much smaller than COMPAS (200 test rows against 1,056), which turned out to be the condition the post-processing comparison depends on.

4.2 Conditions compared

All four conditions use the same predictor architecture, preprocessing and 30-seed protocol, so the debiasing mechanism is the only thing that differs:

1. **Baseline**: predictor only, with no adversary and no debiasing. This shows what the bias looks like unmitigated.
2. **Fixed alpha**: Zhang et al.'s adversarial game at $\alpha \in \{0.5, 1.0, 2.0\}$.
3. **Adaptive** (ours): dual-ascent two-constraint controller, Section 3.4.
4. **Post-processing**: Hardt et al.'s equalized-odds calibration, Section 3.5.

4.3 Statistical methodology

Each condition runs 30 seeds, paired by seed. The train/test split itself is fixed and identical across every seed and condition; only weight initialization and minibatch shuffling vary with the seed. This limits the comparison to training stochasticity, and makes the paired test exact: condition A, seed s and condition B, seed s differ only in the debiasing mechanism and saw the same data. Significance is assessed with the Wilcoxon signed-rank test on paired per-seed values, using the Holm step-down procedure to correct for the multiple comparisons across conditions and metrics (`src/significance_test.py`).

5 Results

5.1 COMPAS

Paired Wilcoxon signed-rank tests, Holm-corrected across all 12 comparisons (full table in `results/significance_compas.txt`):

- **Adaptive vs. best fixed alpha ($\alpha = 2.0$)**: accuracy is not significantly different ($p_{\text{holm}} = 0.087$). DEO is significantly better for adaptive ($p_{\text{holm}} < 0.0001$) and DAO is significantly worse ($p_{\text{holm}} = 0.027$). Adaptive reaches a different point on the frontier automatically, and it does not dominate fixed $\alpha = 2.0$.
- **Post-processing vs. adaptive**: significantly better on all three metrics ($p_{\text{holm}} < 0.001$ for each).

Condition	Accuracy	DEO	DAO
Baseline (no mitigation)	0.657 ± 0.005	0.249 ± 0.012	0.207 ± 0.015
Fixed $\alpha = 0.5$	0.648 ± 0.005	0.244 ± 0.014	0.203 ± 0.020
Fixed $\alpha = 1.0$	0.638 ± 0.010	0.227 ± 0.022	0.192 ± 0.019
Fixed $\alpha = 2.0$	0.615 ± 0.011	0.167 ± 0.023	0.131 ± 0.023
Adaptive (ours)	0.609 ± 0.009	0.127 ± 0.019	0.144 ± 0.014
Post-processing	0.631 ± 0.020	0.061 ± 0.045	0.060 ± 0.040

Table 1: COMPAS, 30 seeds, mean ± std.

- **Adaptive vs. baseline**: a significant accuracy cost ($p_{\text{holm}} = 1.2 \times 10^{-5}$) buys a significant reduction in DEO ($p_{\text{holm}} = 2.2 \times 10^{-8}$) and DAO ($p_{\text{holm}} = 1.2 \times 10^{-5}$).
- **Post-processing vs. baseline**: a smaller significant accuracy cost than adaptive's, with a larger significant DEO/DAO reduction (all $p_{\text{holm}} < 3 \times 10^{-5}$).

Figure 2 plots accuracy against DEO and DAO for every condition. Post-processing has the lowest DEO and DAO of any condition, at an accuracy between fixed $\alpha = 1.0$ and fixed $\alpha = 2.0$. It is below and to the right of the adaptive method on both panels. We did not test post-processing against the fixed-alpha conditions for significance.

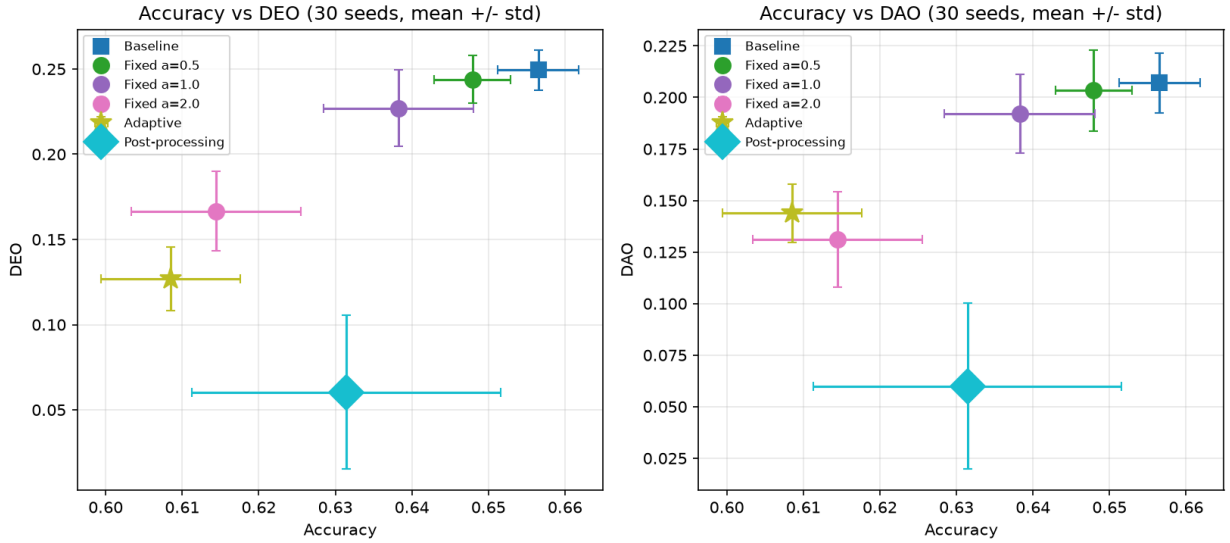


Figure 2: Accuracy vs. DEO/DAO, COMPAS, 30 seeds (mean ± std). Lower DEO/DAO and higher accuracy are better. Post-processing (diamond) has the lowest DEO and DAO of any condition.

5.2 German Credit

The pattern reverses. Post-processing is the worst condition on accuracy and DAO, with roughly double the standard deviation of every other condition on every metric. Paired testing (full table in `results/significance_german.txt`, Holm-corrected across 15 comparisons) confirms the DAO difference: post-processing vs. adaptive, vs. fixed $\alpha = 2.0$ and vs. baseline are all significant ($p_{\text{holm}} =$

Condition	Accuracy	DEO	DAO
Baseline (no mitigation)	0.721 ± 0.016	0.058 ± 0.037	0.053 ± 0.025
Fixed $\alpha = 0.5$	0.726 ± 0.016	0.063 ± 0.039	0.056 ± 0.030
Fixed $\alpha = 1.0$	0.729 ± 0.015	0.045 ± 0.027	0.057 ± 0.027
Fixed $\alpha = 2.0$	0.726 ± 0.015	0.047 ± 0.028	0.053 ± 0.023
Adaptive (ours)	0.722 ± 0.016	0.049 ± 0.036	0.049 ± 0.025
Post-processing	0.711 ± 0.036	0.061 ± 0.055	0.105 ± 0.067

Table 2: German Credit, 30 seeds, mean ± std.

$0.0012, 0.0011, 0.0030$ respectively). The DEO and accuracy differences point the same way but are not statistically significant at $n = 30$ on this smaller dataset. Adaptive vs. baseline shows no significant difference on any metric on German Credit, which fits its low baseline bias leaving little for any debiasing mechanism to fix.

6 Discussion

What adaptive alpha control achieves. On COMPAS, paired significance testing (Wilcoxon, Holm-corrected) shows the dual-ascent controller makes a different trade from the best fixed alpha found by sweeping. Against fixed $\alpha = 2.0$, accuracy is not significantly different (0.609 vs. 0.615, $p_{\text{holm}} = 0.087$), DEO is significantly better (0.127 vs. 0.167, $p_{\text{holm}} < 0.0001$), and DAO is significantly worse (0.144 vs. 0.131, $p_{\text{holm}} = 0.027$). So the controller finds a different point on the frontier automatically, and that point is not strictly better. At this operating point it over-weights the DEO constraint relative to DAO, which follows from the relative convergence rates of the two dual-ascent multipliers. A practitioner who cares most about equal opportunity (DEO) gains from the adaptive mechanism without searching; one who weights DAO more would still need to sweep.

Post-processing wins on COMPAS, but only conditionally. If post-processing beats adversarial training, it is fair to ask why anyone would use adversarial training at all. The German Credit result answers this: post-processing's advantage depended on COMPAS's comparatively large held-out set. Adversarial methods (fixed alpha and adaptive) use the full training set during optimization and do not share this weakness. They stayed stable across both datasets, while post-processing's DAO got significantly worse on the smaller one. In practice, it is worth checking how much held-out data is available for calibration before choosing a debiasing approach on the strength of a COMPAS result like ours.

Why German Credit moved so little. German Credit's baseline DEO/DAO (0.058 / 0.053) are already small compared with COMPAS's (0.249 / 0.207). Gender disparity in this dataset is a smaller problem than racial disparity in COMPAS's recidivism predictions. Every adversarial method tested landed within noise of baseline on German Credit. That says more about the dataset than about the method, because there was little bias to remove.

The ablations did not improve on the mechanism. We tested two alternative designs against the dual-ascent controller. The first annealed ϵ_{DEO} and ϵ_{DAO} from the baseline gap down to their final targets, instead of fixing them from epoch 1. The second replaced the alternating updates with a single joint gradient-reversal backward pass. Annealing gave up some DEO reduction in exchange for accuracy, landing at a different point on the same frontier. Gradient reversal reproduced the alternating-update results almost exactly, so the alternating implementation was not costing anything. Neither beat the adaptive mechanism's own frontier point, which suggests

the frontier itself limits the results more than the training procedure does.

7 Limitations and Ethical Considerations

Test data feeds the controller and the accuracy floors. The experiments have no separate validation split. For the adaptive method, DEO, DAO and accuracy are computed on the test split after every epoch to drive alpha's update, and the same split gives the reported numbers. The accuracy floors were also chosen after looking at the test accuracy that fixed $\alpha = 2.0$ reached. The test split is never used for gradient updates, but these choices let information from it into training, so the adaptive results are likely to be somewhat optimistic. Giving the controller a validation split carved out of the training data would remove this leak, and is the first change we would make in future work.

Binary sensitive attributes. Race in COMPAS (restricted to African-American vs. Caucasian) and gender in German Credit are both treated as binary. This follows how these datasets record the attribute and is not something the method endorses. Neither dataset tests intersectional or non-binary protected groups.

Narrow fairness definitions. DEO and DAO are two equalized-odds-family definitions among many. They do not cover calibration parity, individual fairness, or counterfactual fairness.

The accuracy floor is partly an engineered guarantee. Because the hard clip stops alpha from increasing while accuracy is below a floor, the preserved accuracy is partly designed in and is not purely a property of the dual-ascent mechanism. It should be read as a disclosed design choice.

The controller's own hyperparameters were tuned. η (the ascent step size) and the 30% fraction used to set ϵ_{DEO} and ϵ_{DAO} were chosen by hand, based on what produced a working mechanism within a 30-epoch budget. They were not derived from first principles or found by an independent search. This is the same kind of tuning that produced the fixed-alpha values in the comparison, and the comparison should be weighed with that in mind.

The post-processing finding is conditional and not fully characterized. Two datasets are enough to show that post-processing's win on COMPAS does not generalize unconditionally, and that calibration-set size is the likely cause. They are not enough to find the data size at which post-processing stops being reliable, or to rule out other factors such as class balance, group-size imbalance or the dimension of the feature space. This remains an open question.

8 Conclusion

Adaptive Alpha Control removes the need to hand-sweep a fixed weight in adversarial debiasing, reaching a comparable operating point automatically through dual ascent on two explicit fairness constraints. The larger finding is that a much simpler post-hoc calibration method outperforms it on COMPAS, though only when there is enough calibration data. The choice between in-processing and post-processing debiasing should depend on how much held-out data is available, and a single-dataset benchmark in either direction is not enough to decide it. Future work could locate this threshold more precisely, test whether a hybrid (post-processing when data allows, adversarial training as a fallback when it does not) outperforms either alone, and extend the two-constraint dual-ascent idea to non-binary or intersectional sensitive attributes.

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